

Customer Support Ticket Resolution Pipeline

Problem Statement — Detailed Explanation

1. Background — What Happened?

A company's Operations team ran a two-day Customer Support Efficiency Review. During these two days, support agents resolved customer tickets. Every time an agent finished working on a ticket, a record of that work was saved into a CSV file.

At the end of each day, those CSV files were uploaded to a cloud storage system called Azure Data Lake Storage Gen2 (ADLS Gen2) — think of it as a shared hard drive in the cloud where the company stores all its data files.

Simple analogy: *Imagine a call-centre where each agent keeps a paper log of every customer call they handled. At the end of each shift, those paper logs are scanned and saved to a shared folder. ADLS Gen2 is that shared folder, and each CSV file is one day's worth of scanned logs.*

After two days, leadership wanted answers to specific performance questions. But the raw CSV files are just rows of data — they cannot be queried, filtered, or visualised without first being cleaned and organised. That is the core problem this pipeline solves.

2. The Core Problem

Leadership needs a clean, structured, query-ready dataset that lets them answer four specific business questions. The raw CSV files on ADLS Gen2 cannot answer these questions directly because:

- The data is spread across two separate files (one per day).
- The files contain unresolved tickets, invalid entries, and agents outside the review scope.
- Resolution times are stored as text strings (e.g. "1h 10m 20s") rather than numbers, making comparisons impossible without conversion.
- There is no link between a ticket and the agent's full profile (name, role, team lead).

The four questions leadership needs answered are explained in detail below.

3. The Four Business Questions

Question 1 — Ticket Resolution Rates Across the Team Hierarchy

Each support agent reports to a Team Lead. There are eight Team Leads in scope (TL01 through TL08). Leadership wants to see how many tickets were successfully resolved under each Team Lead so they can compare team performance.

Example: *Team Lead TL03 had 5 agents. Between them, they resolved 42 tickets on Day 1 and 38 on Day 2. TL07 had 6 agents but only resolved 28 tickets across both days. This comparison helps leadership identify which teams are performing well and which need support.*

Question 2 — Per-Agent Performance for Day 1 and Day 2

Beyond team-level totals, leadership wants to see each individual agent’s performance separately for Day 1 and Day 2. This lets them spot agents who improved, declined, or were only active on one day.

Example: Agent A003 resolved 8 tickets on Day 1 but only 3 on Day 2. Agent A017 had 0 successful tickets on Day 1 but resolved 6 on Day 2. These individual trends are invisible in the raw files without proper structuring.

Question 3 — Compliance with the Resolution Quality Threshold

Not every resolved ticket counts as a genuine resolution. The company has a quality rule: a ticket is only counted as successfully resolved if the agent spent more than 15 minutes working on it. Tickets closed in 15 minutes or less are considered rushed or improperly handled and must be excluded.

Example: Agent A011 closed a ticket in 12 minutes and marked it “Resolved”. Under the quality threshold rule, this ticket does NOT count. Another ticket closed by the same agent in 22 minutes DOES count. Leadership wants to know how many agents and teams are meeting this standard.

How Resolution Time is Stored

Resolution times in the CSV files are stored as text in the format Xh Xm Xs — for example, “0h 22m 45s” means 22 minutes and 45 seconds. Before any comparison can be made, this text must be converted to a plain number (total minutes).

There is also a rounding rule: if the seconds value is 30 or more, the time is rounded up to the next whole minute.

Raw Time String	Converted (decimal minutes)	Counts as Resolved? (threshold > 15 min)
0h 22m 45s	22 + 1 = 23 mins (45s ≥ 30, round up)	✓ Yes
0h 14m 20s	14 mins (20s < 30, no rounding)	✗ No (14 ≤ 15)
1h 10m 30s	70 + 1 = 71 mins (30s ≥ 30, round up)	✓ Yes
0h 15m 00s	15 mins exactly	✗ No (must be > 15, not equal)

Question 4 — Identifying Agents Who Carried Over Unresolved Work from Day 1

If an agent successfully resolved tickets on Day 1, they should not appear in the Day 2 results. The assumption is that a successful Day 1 agent’s work is complete. Only agents who did NOT succeed on Day 1 are expected to continue and be measured on Day 2.

This is called the Day 2 Carry-over Rule. Without this filter, an agent who succeeded on both days would be double-counted, which would inflate team performance numbers.

Example: Agent A005 successfully resolved tickets on Day 1 and is in the Day 1 results. On Day 2, this agent also resolved tickets. Because they already succeeded on Day 1, their Day 2 records are excluded. Agent A009 did NOT succeed on Day 1, so their Day 2 records are kept and counted normally.

4. Scope Constraint — Who Is Included?

Not all agents in the CSV files are part of this review. Only agents who report to Team Leads TL01 through TL08 are included. Any agent belonging to a different team lead is ignored, even if their tickets appear in the data.

Example: Agent A044 reports to TL12, which is outside the review scope. Even if A044 resolved 20 tickets with excellent times, all their records are filtered out. Only agents under TL01–TL08 count.

5. Business Rules Summary

The table below summarises all the rules that define what counts as a valid, successful resolution in this review.

Rule	Definition	Example
Successful Resolution	Status must be “Resolved” AND resolution time must be more than 15 minutes.	Status = Resolved, time = 18 min ✓ Status = Resolved, time = 13 min X
Resolution Time Format	Times are stored as text: Xh Xm Xs. They must be converted to total decimal minutes before any comparison.	“0h 18m 00s” → 18 minutes
Rounding Rule	If seconds are 30 or more, round up to the next full minute. If less than 30, drop the seconds.	22m 45s → 23 min 14m 20s → 14 min
Day 2 Carry-over Rule	Agents who already succeeded on Day 1 are excluded from Day 2 results to avoid double-counting.	A005 succeeded on Day 1 → excluded from Day 2 count
Scope Filter	Only agents under Team Leads TL01–TL08 are included. All others are excluded.	Agent under TL12 → excluded from all results

6. Why a Pipeline Is Needed

All of the above business questions and rules cannot be answered just by opening the CSV files in a spreadsheet. The data needs to be:

1. Ingested — read from cloud storage in a reliable, repeatable way.
2. Validated — checked for missing values, bad formats, and duplicate records.
3. Transformed — resolution times converted from text to numbers, agents joined with their profile data.
4. Filtered — only valid, in-scope, quality-passing records kept.
5. Aggregated — summarised into KPIs per Team Lead and per day, ready for leadership dashboards.

In short: *The pipeline turns messy, raw CSV files into a clean, trustworthy dataset that leadership can use with confidence to make decisions about their customer support team.*